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Roll No

IT-6004-CBGS

B.E. VI Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Compiler Design

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Assume missing data if any, suitably.

1. a) Compare the features of a single pass compiler with multi-pass compiler. 7
- b) Briefly explain the compiler construction tools. 7

2. a) Consider the grammar: 7

$$E \rightarrow 5 + T \mid 3 - T,$$

$$T \rightarrow V \mid V * V \mid V + V,$$

$$V \rightarrow a \mid b.$$

What is the use of left factoring? Find the first and follow for above grammar and do the left factoring for it.

- b) Describe the following in brief. 7
 - i) Ambiguity
 - ii) Left recursion
 - iii) Parse Tree

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3. a) Consider the following grammar 7
 $E \rightarrow E+T$
 $T \rightarrow TF/F$
 $F \rightarrow F*/a/b$
Construct the SLR Parsing table for this grammar.
- b) Construct the predictive parser for the following grammar: 7
 $S \rightarrow (L)/a$
 $L \rightarrow L,S/S$
4. a) Write the grammar and syntax directed definitions for a simple desk calculator and show annotated parse tree for the expression: $(3+4) * (5+6)$. 7
- b) With an example, explain the various formats of intermediate code. 7
5. a) What are the features and capabilities of the symbol table? 7
- b) What is an activation record? Explain the purpose of different fields in an activation record. 7
6. a) Write an algorithm to construct a DAG from a basic block. 7
- b) Generate the three-address code for the following code segment:
While $(a < c \text{ and } b < d)$ do
If $a = 1$ then $c = c + 1$;
Else
While $(a \leq d)$ do
 $a = a + 3$; 7

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Contd...

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7. a) Explain the common sub expression elimination, copy propagation and transformation for moving loop invariant computations in details. 7
- b) Explain the peephole optimization. 7
8. Write short notes on following.(any three) 14
- a) Lexical analysis
 - b) Canonical items
 - c) Back patching
 - d) L-attributes and S-attributes
